Type MS · Exact values with the addition formulae

Answer 1 of 4 · Question 9 part a · 3 marks

2024 JUN · Paper 02 · extract 1 of 4 · find part a in this mark scheme extract

9(a)	$\sin 75^\circ = \sin\left(45^\circ + 30^\circ\right) = \sin 45^\circ \cos 30^\circ + \sin 30^\circ \cos 45^\circ$ $= \frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{\sqrt{2}}{2} = \frac{\sqrt{6}+\sqrt{2}}{4} *$ **ALT 1** $\sin 75^\circ = \sin\left(120^\circ - 45^\circ\right) = \sin 120^\circ \cos 45^\circ - \sin 45^\circ \cos 120^\circ$ $= \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \times \left(-\frac{1}{2}\right) = \frac{\sqrt{6}+\sqrt{2}}{4} *$ **ALT 2** $\sin 75^\circ = \sin\left(60^\circ + 15^\circ\right) = \sin 60^\circ \cos 15^\circ + \sin 15^\circ \cos 60^\circ$ $= \frac{\sqrt{3}}{2} \times \left(\frac{\sqrt{6}+\sqrt{2}}{4}\right) + \left(\frac{\sqrt{6}-\sqrt{2}}{4}\right) \times \frac{1}{2}$ $= \frac{3\sqrt{2}+\sqrt{6}}{8} + \frac{\sqrt{6}-\sqrt{2}}{8} = \frac{\sqrt{6}+\sqrt{2}}{4}$	M1 M1A1 cso [3] [M1 M1A1] [M1 M1 A1]
(b)	$\left[\angle AFC = 180^\circ - 60^\circ - 45^\circ = 75^\circ\right]$ $\frac{24}{\sin 75^\circ} = \frac{BF}{\sin 60^\circ}$ $\Rightarrow \frac{24}{\sin 75^\circ} = \frac{BF}{\sin 60^\circ} \Rightarrow \frac{24}{\frac{\sqrt{6}+\sqrt{2}}{4}} = \frac{BF}{\frac{\sqrt{3}}{2}}, \Rightarrow BF = \frac{48\sqrt{3}}{\sqrt{6}+\sqrt{2}}$ $BF = \frac{48\sqrt{3}}{\left(\sqrt{6}+\sqrt{2}\right)} \times \frac{\left(\sqrt{6}-\sqrt{2}\right)}{\left(\sqrt{6}-\sqrt{2}\right)} = \frac{48\sqrt{3}\sqrt{6}-48\sqrt{6}}{6-2} = 12\left(3\sqrt{2}-\sqrt{6}\right) *$	M1,M1A1 M1A1 cso [5]
(c)	Let the vertical point from F to the line AB be X and the corresponding point on CD be Y. The required angle is EXY $FX = 12\left(3\sqrt{2}-\sqrt{6}\right)\sin 45^\circ = 36 - 12\sqrt{3} = (15.2153...)$ OR Area $ABC = \frac{1}{2} \times 24 \times FX = \frac{1}{2} \times 24 \times 12\left(3\sqrt{2}-\sqrt{6}\right)\sin 45$ $\Rightarrow FX = 12\left(3\sqrt{2}-\sqrt{6}\right)\sin 45^\circ = 36 - 12\sqrt{3} = (15.2153...)$ $EF = \frac{36-12\sqrt{3}}{\tan 65^\circ} = 7.09505.... \approx 7.1 \text{ (cm)}$	M1 M1A1 [3]
(d)	Identifies the required angle FCX $FC = \sqrt{\left[12\left(3\sqrt{2}-\sqrt{6}\right)\right]^2 + '7.09505'^2} = 22.65735.....$ $\angle FCX = \sin^{-1}\left(\frac{15.2153}{'22.6573'}\right) = 42.186...^\circ \approx 42.2^\circ$	B1 M1 M1A1 [4]
		Total 15 marks

Type MS · Exact values with the addition formulae

Answer 1 of 4 · Question 9 part a · 3 marks

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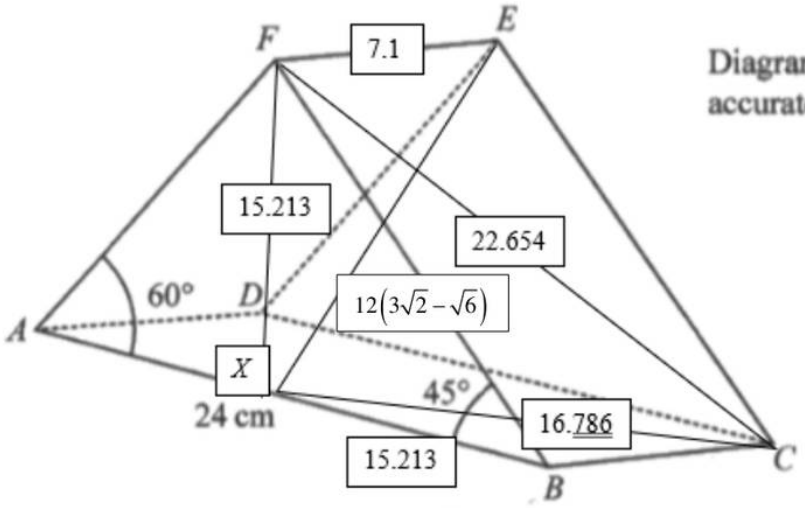
Part	Mark	Notes
(a)	M1	For the correct use of the addition formula for sine with $\sin(45^\circ+30^\circ)$ Accept also $\sin(120^\circ-45^\circ)$ and $\sin(60^\circ+15^\circ)$ **but not** $\sin(90^\circ-15^\circ)$ or $\sin(180^\circ-105^\circ)$
	M1	For applying the exact values of these sines and cosines and for evaluating the expression.. This step must be explicitly seen.
	A1 cso	For the correct final answer with no errors seen
(b)	M1	For the correct use of sine rule using the exact value for sin 75° and sin 60°
	M1	For rearranging to obtain $BF=\frac{k\sqrt{3}}{\sqrt{6}+\sqrt{2}}$ or $\frac{l\sqrt{3}}{2\sqrt{6}+2\sqrt{2}}$ Where k and l are integers
	A1	For the correct **exact** BF which need not be simplified. $BF=\frac{48\sqrt{3}}{\sqrt{6}+\sqrt{2}}$ or $\frac{96\sqrt{3}}{2\sqrt{6}+2\sqrt{2}}$ or even $\frac{12\sqrt{3}\times4}{\sqrt{6}+\sqrt{2}}$ but not for example, $\frac{\frac{12\sqrt{3}}{\sqrt{6}+\sqrt{2}}}{4}$ unless recovery is seen later
	M1	For multiplying numerator **and** denominator by their conjugate which must of the form e.g $\sqrt{A}\pm\sqrt{B}$ **AND** simplifying to the required form. We need to see at least one intermediate line of working before the final answer here.
	A1 cso	The answer as shown with no errors or omissions seen. This is a given result.
(c)	M1	For finding the length FX using any appropriate trigonometry.
	M1	For using any appropriate trigonometry to find length EF [or XY]
	A1	For awrt 7.1 (cm)
(d)	B1	For identifying the correct angle required.
	M1	For using Pythagoras theorem to find FC $FC=\sqrt{21.51781^2+7.09505^2}=22.65735.....$ Or For using Pythagoras theorem to find CX $FC=\sqrt{15.213^2+7.095^2}=16.786...$ [Note: $BX=15.213...$ and $CX=16.786\ldots$]
	M1	For using any appropriate trigonometry to find angle FCX. For example; $\angle FCX=\tan^{-1}\left(\frac{15.2153}{16.786}\right)=42.186...^\circ\approx42.2^\circ$
	A1	For awrt 42.2°

Type MS · Exact values with the addition formulae

Answer 1 of 4 · Question 9 part a · 3 marks

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USEFUL SKETCH



NB:

$AC = 25.03$ cm

$AF = 17.57$ cm

$CF = 22.65$ cm

Angle $FCA = 42.9°$

Type MS · Exact values with the addition formulae

Answer 1 of 4 · Question 9 part a · 3 marks

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Question	Scheme	Marks

Type MS · Exact values with the addition formulae

Answer 2 of 4 · Question 2 part b · 3 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

M1	Uses b^2-4ac on the given quadratic equation with correct a, b and c; $a=3(k+2)$ $b=k+5$ $c=k$ **and** a correct substitution to obtain $(k+5)^2-4\times3\times(k+2)(k)$ **Note:** Accept for this mark any inequality, equals sign and even b^2-4ac used on its own.
M1	For attempting to expand the brackets and form a 3TQ in terms of k. Allow as a minimum at least one term correct. $k^2+10k+25-12k^2-24k \Rightarrow (-11k^2-14k+25)$ M0M1 is possible here.
A1	For the correct 3TQ with the correct inequality. **Note:** Allow > or < in place of $\geqslant$ and $\leqslant$ for this mark $-11k^2-14k+25 \geqslant 0$ or $11k^2+14k-25 \leqslant 0$
M1	For an attempt to solve their 3TQ, (provided it is a 3TQ) in terms of k by any acceptable method. See General Guidance for the definition of an attempt by factorisation, formula or completing the square. Use of calculators: if their 3TQ is incorrect, do not award this mark if working is not seen. $(11k+25)(k-1)=0 \Rightarrow k=1,\ -\frac{25}{11}$
M1	For forming the correct inequality with **their** critical values, provided they have been obtained from a 3TQ, must be a closed region. $\left(-\frac{25}{11} \leqslant k \leqslant 1\right)$ ft their values from their $-11k^2+14k-25 \geqslant 0$ or $11k^2-14k+25 \leqslant 0$
A1	For the correct inequality. $-\frac{25}{11} \leqslant k \leqslant 1$

Type MS · Exact values with the addition formulae

Answer 2 of 4 · Question 2 part b · 3 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks
2 (a)(i)	$(\tan\alpha =)\dfrac{4}{3}$	B1
(a)(ii)	$(\tan\beta =)-\dfrac{1}{\sqrt{3}}\quad\left(=-\dfrac{\sqrt{3}}{3}\right)$	B1 B1 (3)
(b)	$\tan(\alpha+\beta)=\dfrac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta}$	
	$=\dfrac{\frac{4}{3}+\left(-\frac{1}{\sqrt{3}}\right)}{1-\left(\frac{4}{3}\right)\left(-\frac{1}{\sqrt{3}}\right)}$	M1
	$=\dfrac{\frac{4\sqrt{3}\pm 3}{3\sqrt{3}}}{\frac{3\sqrt{3}\pm 4}{3\sqrt{3}}}$ or $\dfrac{\frac{12+\left(-3\sqrt{3}\right)}{9}}{\frac{9+4\sqrt{3}}{9}}$	dM1
	$=\dfrac{4\sqrt{3}-3}{3\sqrt{3}+4}$	A1 cso (3)
		Total 6 marks

Type MS · Exact values with the addition formulae

Answer 2 of 4 · Question 2 part b · 3 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Exact values with the addition formulae

Answer 3 of 4 · Question 11 part c.i · 2 marks

2020 JAN · Paper 02 · extract 1 of 4 · find part c.i in this mark scheme extract

11(a)	$(3\sin A\cos B - 3\cos A\sin B) = (\sin A\cos B + \cos A\sin B)$	M1
	$2\sin A\cos B = 4\cos A\sin B$	M1
	$\rightarrow \frac{\sin A}{\cos A} = 2\frac{\sin B}{\cos B}$	M1
	$\tan A = 2\tan B \quad k = 2$	A1 (4)
(b)	$\frac{(\cos^4\theta - \sin^4\theta)}{\cos^2\theta} = \frac{(\cos^2\theta + \sin^2\theta)(\cos^2\theta - \sin^2\theta)}{\cos^2\theta}$	M1
	$= \frac{(\cos^2\theta - \sin^2\theta)}{\cos^2\theta}$	M1
	$= 1 - \tan^2\theta$ *	A1 cso (3)
ALT 1	$1-\tan^2\theta = 1-\frac{\sin^2\theta}{\cos^2\theta} = \frac{\cos^2\theta-\sin^2\theta}{\cos^2\theta}$	M1
	$= \frac{\cos^2\theta-\sin^2\theta}{\cos^2\theta}\times\left(\cos^2\theta+\sin^2\theta\right)$	M1
	$= \frac{\cos^4\theta-\sin^4\theta}{\cos^2\theta}$	A1
ALT 2	$\frac{\cos^4\theta-\sin^4\theta}{\cos^2\theta} = \cos^2\theta - \frac{\sin^4\theta}{\cos^2\theta} = \cos^2\theta - \tan^2\theta\sin^2\theta$	M1 Eliminate 4th powers
	$= \cos^2\theta - \tan^2\theta\left(1-\cos^2\theta\right)$	M1 Eliminate $\sin^2$
	$= \cos^2\theta - \tan^2\theta + \sin^2\theta = 1-\tan^2\theta$	A1
(c)(i)	$\cos(45-30)$ or $\cos(60-45) = \frac{1}{2}\times\frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2}\times\frac{\sqrt{2}}{2}$	M1
	$= \frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4} = \frac{\sqrt{2}+\sqrt{6}}{4}$ *	A1cso (2)
ALT	By using double angle formula: $\cos^2 15° = \frac{1}{2}\left(1+\cos 30°\right) = \frac{1}{2}\left(1+\frac{\sqrt{3}}{2}\right)$	M1
	Leading to the *given* answer. $\cos 15° = \sqrt{\left(\frac{2+\sqrt{3}}{4}\right)}$ or $\frac{\sqrt{2+\sqrt{3}}}{2}$ must be seen.	A1

Type MS · Exact values with the addition formulae

Answer 3 of 4 · Question 11 part c.i · 2 marks

2020 JAN · Paper 02 · extract 2 of 4 · find part c.i in this mark scheme extract

Question Number	Scheme	Marks
(ii)	$\tan 255 = \tan 75$	B1
	$= \tan(30 + 45) = \dfrac{\tan 30 + \tan 45}{1 - \tan 30 \tan 45}$	M1
	$= \dfrac{\frac{\sqrt{3}}{3}+1}{1-\frac{\sqrt{3}}{3}}$	dM1
	$\dfrac{\frac{3+\sqrt{3}}{3}}{\frac{3-\sqrt{3}}{3}} = \dfrac{3+\sqrt{3}}{3-\sqrt{3}}$ *	A1cso (4)
		[13]

(a)

M1 Expand both sides of the equation using correct formulae
M1 Collect like terms from their expansions. (Not dependent)
M1 Divide through by $\cos A \cos B$
A1 Replace each fraction with the appropriate tangent and show $k = 2$ (value need not be shown explicitly)

(b)

M1 Factorise the numerator using the difference of 2 squares.
M1 Replace $\sin^2\theta + \cos^2\theta$ with 1
A1cso Divide both terms by $\cos^2\theta$ and obtain the *given* answer with no errors seen.

ALT 1

M1 Use $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$ and obtain a single fraction with no tan
M1 Indicate multiplication by $\sin^2\theta + \cos^2\theta$
A1cso Multiply and obtain the *given* answer with no errors seen.

(c)(i)

M1 Express 15 as the difference of 2 suitable numbers, expand using a correct formula and substitute the correct exact values for the trig functions (substitution must be shown).
A1cso Simplify and combine the fractions to obtain the *given* answer with no errors seen.

(ii)

B1 $\tan 255 = \tan 75$ seen explicitly or used.
OR eg tan(210 + 45) – give B1 for tan 210 = tan 30 used
M1 Express 75 as 30 + 45 and expand $\tan(30+45)$ using the correct formula (given on the formula page) OR expand tan (210 + 45)
If 75 = 15 + 60 is used tan 15 can be obtained from a calculator but must be in exact form..
dM1 Substitute the correct exact values for the trig functions
A1cso Simplify the fractions to obtain the *given* answer with full working and no errors seen.

Type MS · Exact values with the addition formulae

Answer 3 of 4 · Question 11 part c.i · 2 marks

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Type MS · Exact values with the addition formulae

Answer 3 of 4 · Question 11 part c.i · 2 marks

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Type MS · Exact values with the addition formulae

Answer 4 of 4 · Question 11 part c.ii · 4 marks

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11(a)	$(3\sin A\cos B - 3\cos A\sin B) = (\sin A\cos B + \cos A\sin B)$	M1
	$2\sin A\cos B = 4\cos A\sin B$	M1
	$\rightarrow \frac{\sin A}{\cos A} = 2\frac{\sin B}{\cos B}$	M1
	$\tan A = 2\tan B \quad k = 2$	A1 (4)
(b)	$\frac{(\cos^4\theta - \sin^4\theta)}{\cos^2\theta} = \frac{(\cos^2\theta + \sin^2\theta)(\cos^2\theta - \sin^2\theta)}{\cos^2\theta}$	M1
	$= \frac{(\cos^2\theta - \sin^2\theta)}{\cos^2\theta}$	M1
	$= 1 - \tan^2\theta$ *	A1 cso (3)
ALT 1	$1-\tan^2\theta = 1-\frac{\sin^2\theta}{\cos^2\theta} = \frac{\cos^2\theta-\sin^2\theta}{\cos^2\theta}$	M1
	$= \frac{\cos^2\theta-\sin^2\theta}{\cos^2\theta}\times\left(\cos^2\theta+\sin^2\theta\right)$	M1
	$= \frac{\cos^4\theta-\sin^4\theta}{\cos^2\theta}$	A1
ALT 2	$\frac{\cos^4\theta-\sin^4\theta}{\cos^2\theta} = \cos^2\theta - \frac{\sin^4\theta}{\cos^2\theta} = \cos^2\theta - \tan^2\theta\sin^2\theta$	M1 Eliminate 4th powers
	$= \cos^2\theta - \tan^2\theta\left(1-\cos^2\theta\right)$	M1 Eliminate $\sin^2$
	$= \cos^2\theta - \tan^2\theta + \sin^2\theta = 1-\tan^2\theta$	A1
(c)(i)	$\cos(45-30)$ or $\cos(60-45) = \frac{1}{2}\times\frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2}\times\frac{\sqrt{2}}{2}$	M1
	$= \frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4} = \frac{\sqrt{2}+\sqrt{6}}{4}$ *	A1cso (2)
ALT	By using double angle formula: $\cos^2 15° = \frac{1}{2}\left(1+\cos 30°\right) = \frac{1}{2}\left(1+\frac{\sqrt{3}}{2}\right)$	M1
	Leading to the *given* answer. $\cos 15° = \sqrt{\left(\frac{2+\sqrt{3}}{4}\right)}$ or $\frac{\sqrt{2+\sqrt{3}}}{2}$ must be seen.	A1

Type MS · Exact values with the addition formulae

Answer 4 of 4 · Question 11 part c.ii · 4 marks

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Question Number	Scheme	Marks
(ii)	$\tan 255 = \tan 75$	B1
	$= \tan(30 + 45) = \dfrac{\tan 30 + \tan 45}{1 - \tan 30 \tan 45}$	M1
	$= \dfrac{\frac{\sqrt{3}}{3}+1}{1-\frac{\sqrt{3}}{3}}$	dM1
	$\dfrac{\frac{3+\sqrt{3}}{3}}{\frac{3-\sqrt{3}}{3}} = \dfrac{3+\sqrt{3}}{3-\sqrt{3}}$ *	A1cso (4)
		[13]

(a)

M1 Expand both sides of the equation using correct formulae
M1 Collect like terms from their expansions. (Not dependent)
M1 Divide through by $\cos A \cos B$
A1 Replace each fraction with the appropriate tangent and show $k = 2$ (value need not be shown explicitly)

(b)

M1 Factorise the numerator using the difference of 2 squares.
M1 Replace $\sin^2\theta + \cos^2\theta$ with 1
A1cso Divide both terms by $\cos^2\theta$ and obtain the *given* answer with no errors seen.

ALT 1

M1 Use $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$ and obtain a single fraction with no tan
M1 Indicate multiplication by $\sin^2\theta + \cos^2\theta$
A1cso Multiply and obtain the *given* answer with no errors seen.

(c)(i)

M1 Express 15 as the difference of 2 suitable numbers, expand using a correct formula and substitute the correct exact values for the trig functions (substitution must be shown).
A1cso Simplify and combine the fractions to obtain the *given* answer with no errors seen.

(ii)

B1 $\tan 255 = \tan 75$ seen explicitly or used.
OR eg tan(210 + 45) – give B1 for tan 210 = tan 30 used
M1 Express 75 as 30 + 45 and expand $\tan(30+45)$ using the correct formula (given on the formula page) OR expand tan (210 + 45)
If 75 = 15 + 60 is used tan 15 can be obtained from a calculator but must be in exact form..
dM1 Substitute the correct exact values for the trig functions
A1cso Simplify the fractions to obtain the *given* answer with full working and no errors seen.

Type MS · Exact values with the addition formulae

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